

Research paper

Efficacy of sub-MIC level of meropenem and ciprofloxacin against extensive drug-resistant (XDR) *Pseudomonas aeruginosa* isolates of diabetic foot ulcer patients

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ABSTRACT

The increasing emergence of extensive drug-resistant bacteria (XDR) among chronic diabetic foot ulcer patients (DFU) possess serious threat which leads to foot amputation. The ideal measurement estimations of the presently accessible medications are getting insufficient against extensive drug-resistant strains. For quite a long-time piperacillin monotherapy, Piperacillin-tazobactam, ceftazidime, Carbapenem class of anti-toxin, ceftazone-Tazobactam, and so on, has been the suggested therapy towards persistent instances of diabetic foot ulcer but because of the resistance mechanism of the potent pathogens the potency and usage of the antibiotic concentration regime is under the radar. Based on this hypothesis two isolates namely VIT PC 7 & VIT PC 9 were found to be resistant to all five classes of antibiotics exhibiting extensive drug resistance (XDR). The whole-genome sequence of *Pseudomonas aeruginosa* VIT PC 7 and VIT PC 9 data showed the presence of various RND efflux related genes and antibiotic resistance genes. The broth microdilution assay showed minimum inhibitory concentration (MIC) for ciprofloxacin and meropenem, Synergistic test was performed through checkerboard analysis and sub-MIC concentration of ciprofloxacin/meropenem was deduced using $\sum$ FICI, Time kills analysis was done for varying time interval to check the maximum reduction in CFU/ml of the bacterial cells, sub-MIC level of meropenem and ciprofloxacin showed inhibitory activity at lower concentration respectively. In vitro time-kill analysis showed the decrease in the number of cells, suggesting that the synergistic antimicrobial combinations are effective in decreasing the MIC level, and combinational test involving sub-MIC level of antibiotics also showed maximum reduction in biofilm forming cells, portraying the effectiveness of both the drugs. Accordingly, an expansion in the antimicrobial spectrum can be accomplished by utilizing the ideal measurements of ciprofloxacin/meropenem in persistent condition like diabetic foot ulcer, sub-MIC level of ciprofloxacin/meropenem could be a promising choice for anticipation against the ongoing drug-resistant crisis.

1. Introduction

Diabetic foot ulcer is a chronic condition of diabetic patients (Bjarnsholt et al., 2008), which includes both Type 1 and Type 2 diabetes (Leahy, 2005). The medical trauma includes various factors for chronic condition i.e., disruption of the vascular system, peripheral neuropathy, and invasion of microbes (Bowering, 2001) which leads to lower limb amputation, osteomyelitis (Lipsky, 2014), polymicrobial infection and disruption of blood glucose vessel provides the route for various opportunistic pathogen which worsen the healing condition of the wound.

Pseudomonas aeruginosa and *Staphylococcus* are most common type of

opportunistic pathogen which are routinely screened from chronic wound regions, but the intensity of infection varies from the location of microbiota (Parvez et al., 2012). Hence in more chronic condition presence of *pseudomonas* makes it more potent (Rahim et al., 2017), *Pseudomonas aeruginosa* infection results in delaying of wound healing because of biofilm formation and also lower region amputation (Ertugrul et al., 2017). Biofilm formation by *pseudomonas* sp in case of diabetic foot ulcer prevents angiogenesis and often creates a barrier between drug and the local environment (Oates et al., 2014). Delayed wound healing has also been associated with biofilm formation (Zhao et al., 2010) and in the case of foot ulcers affected by *pseudomonas*, the effect of tropical drugs or antibiotics generally stops because of its

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inability to overcome the barrier. Extensive drug-resistant *Pseudomonas aeruginosa* has been observed in the case of diabetic foot ulcers as compared to non-diabetic wounds (Trivedi et al., 2014) and also it has been associated with polymicrobial infection (Vatan et al., 2018) (Pastor et al., 2013).

Pseudomonas aeruginosa has been routinely associated with diabetic foot ulcer and the presence of class I integron beta-lactamase make it less prone to antibiotic unlike extensive drug-resistant (Shahi and Kumar, 2016; Qudus et al., 2019). Colistin therapy in combination with rifampicin and imipenem has been used against multiple drug-resistant infections in the recent past (Tasini et al., 2006), but carbapenem resistance has been seen in the past few years (Sreedharan et al., 2018). Fluoroquinolones like ciprofloxacin and colistin combination have successfully been used in the case of multi-drug-resistant bacteria (D'Souza et al., 2014). Ceftolozane (new generation cephalosporin) has been tested in the case of extended beta-lactamase-producing *pseudomonas* and also showed potent results against infection (Melchers et al., 2017). Ceftolozane/Tazobactam combination has been studied in soft tissue infection and showed some promising results against extensively resistant *pseudomonas* wound infection (Dietl et al., 2018) though these combinations have some serious side effects. Seeing some other alternatives like Piperacillin/Tazobactam (Pip/Taz) combination has been successfully administered against diabetic foot infection and which showed (Yim et al., 2016) potent results when the prolonged infusion of Pip/Taz have shown some benefit against *pseudomonas* infection (Zelenitsky et al., 2016) which shows some promising results against resistant strains.

Carbapenem class of antibiotics have shown to possess a beneficial role against soft tissue skin infections (Douglas, 2006), Meropenem has shown some excellent results when it comes to gram-negative and gram-positive infections (Odenholt et al., 1997), and has been continuously used against serious *pseudomonas* associated infections, Meropenem-Ceftazidime combinations had been proved to be efficient against extensive care unit pathogens (Feng et al., 2017) and carbapenem class of antibiotics was potent against ventilator-associated pneumonia (Luyt et al., 2014). Fluoroquinolone/meropenem combination has been studied against the hypermutable strain of *pseudomonas* infection during cystic fibrosis, the time-kill analysis till 72 h yielded a synergistic method in combating against hypervirulent strains of *pseudomonas* (Rees et al., 2018), other studies (American and Hoşgör, 2001) conducted on *pseudomonas* and *acinetobacter* infection showed a synergistic effect of meropenem with ciprofloxacin and colistin, the time-kill analysis showed the inhibitory effect of sub-MIC level of meropenem and ciprofloxacin or colistin. A study conducted on nosocomial infection by multidrug-resistant *Pseudomonas aeruginosa* (Isenberg et al., 1999) showed a synergistic effect of meropenem/ciprofloxacin combination and ciprofloxacin/imipenem combination, meropenem/ciprofloxacin (Hidayat et al., 2006; Idowu et al., 2019) have also shown synergistic effect against ICU related *pseudomonas* infection, the time-kill analysis showed the increase in resistant pattern at 0.5X MIC and 1 × MIC. Based on these hypotheses, in our current study we have investigated the synergistic and bactericidal effect of meropenem alone and in combination with ciprofloxacin against biofilm-forming extensive drug-resistant *Pseudomonas aeruginosa* isolated from chronic diabetic foot ulcer condition.

2. Materials and methods

2.1. Characterization of *Pseudomonas aeruginosa* isolated from a diabetic foot ulcer

Nine positive chronic diabetic foot ulcer derived *P. aeruginosa* strains were isolated from the pus region of the diabetic wound from Government Vellore Medical College, Vellore Tamil Nadu. All isolates were identified using 16S rRNA sequencing and NCBI accession number was given, biochemical identification was also performed for all the isolates

Pseudomonas PAO1 was used throughout this study as a reference strain.

2.2. Culture media and antibiotics used

Three different types of media were used to check the growth of clinical isolates. Luria Bertani broth (LB) (HIMEDIA), Brain Heart Infusion broth (BHI) (HIMEDIA), and M63 minimal media were prepared and growth was monitored for 24 h at 37 °C in shaking condition. Further for identification of *pseudomonas sp* King's B medium (HIMEDIA) was used. Glycerol stocks were also prepared for the clinical isolates which include 40% glycerol diluted with 2 × BHI broth and was stored at −80 °C till further use. The antipseudomonal antibiotics used in this experiment were piperacillin, ciprofloxacin, gentamicin, ceftazidime obtained from Himedia Laboratories, Tazobactam sodium salt was obtained from Sigma Aldrich. Antibiotic solutions were prepared according to the Clinical and Laboratory Standards Institute (CLSI) guidelines and stored frozen at −80 °C for up to 6 months, drug diffusion discs were purchased from Himedia Laboratories. Antibiotic breakpoints for the interpretative criteria for clinical isolates of *pseudomonas sp* was used according to the CLSI guidelines.

2.3. Biofilm detection assay

Biofilm development was evaluated using the plate calorimetric method, Bacterial colonies were grown in Tryptic soy broth (TSB) (HIMEDIA) medium for 24 h, The culture was diluted (1:100) in a new TSB broth, from that 2 ml of culture was pipetted out in small test tubes, The tubes were then incubated for 24 h at 37 °C without shaking, The tubes were then gently washed with an equal volume of 1 × phosphate-buffered saline, after washing the tubes were then kept for air drying, 0.1% (w/v) crystal violet (CV) was then added to the tubes and were incubated for 10 min at room temperature. The protocol was done in triplicate and the mean value was considered for further analysis. Isolates that formed biofilm more than the standard isolate were considered as strong biofilm producers and the rest were called weak biofilm producers. Subsequently, SEM analysis was performed by growing VIT PC 7 and VIT PC 9 on glass coverslip for 24 h, the slips were then washed and were fixed with 30% glutaraldehyde and were then analyzed.

2.4. Minimum inhibitory concentration (MIC)

MICs were determined by the broth microdilution techniques as described by CLSI. Serial dilutions ranging from 1024 to 1 µg/ml for piperacillin, piperacillin-tazobactam, ciprofloxacin, meropenem, gentamicin, and ceftazidime were prepared in cation adjusted Muller Hinton broth (CAMHB) microtiter plates. The culture broth was grown that gave a final concentration of 5×10^5 CFU/ml; the plates were then incubated at 37 °C for 18–20 h. The MIC is defined as the minimum antibiotic combination giving invisible growth. CLSI criteria and MIC breakpoints for *Pseudomonas* were recommended as piperacillin (≤16 µg (susceptible), 32–64 µg (intermediate), ≥128 µg (resistant)), for ciprofloxacin (≤1 µg (susceptible), 2 µg (intermediate), ≥4 µg (resistant)), for meropenem (≤2 µg (susceptible), 4 µg (intermediate), ≥8 µg (resistant)) for ceftazidime (≤8 µg (susceptible), 16 µg (intermediate), ≥32 µg (resistant)), and for gentamicin (≤4 µg (susceptible), 8 µg (intermediate), ≥16 µg (resistant)). *Pseudomonas aeruginosa* PAO1 was used as a control strain for all the experiments.

2.5. Whole-genome sequence analysis of extensive drug-resistant isolates

The sequencing library was prepared according to the Oxford Nanopore's SQK-LSK108 kit protocol and was loaded onto a flow cell (R9.4 chemistry) and ran for 12 h on Oxford Nanopore's Minion platform. Base-calling was done using Guppy base caller v1.6.0, and assembly was performed de novo using Flye assembler version 2.6, The sequences were annotated by a comprehensive antibiotic resistance database (CARD)

Table 1Biochemical tests for *Pseudomonas* clinical isolates.

Strain	Oxidase	Catalase	Fluorescein	Methyl Red	Voges Proskauer	Protease production
CL 3	+	+	+	–	–	+
CL 4	+	+	+	–	–	+
CL 5	+	+	+	–	–	+
CL 6	+	+	+	–	–	+
CL 7 (VIT PC 7)	+	+	+	–	–	+
CL 8	+	+	+	–	–	+
CL 9 (VIT PC 9)	+	+	+	–	–	+
CL 10	+	+	+	–	–	+
CL 11	+	+	+	–	–	+
PAO1	+	+	+	–	–	+

identified the acquired antibiotic resistance genes, and also the AMR family present in the genome of extensive drug-resistant isolates.

2.6. Checkerboard analysis

Checkerboard studies were performed in 96 well microtiter plates, the antibiotics to be tested includes broad values, the values of the expected MIC were MIC₁/2 of it; 50 µl CAMHB was distributed into each microtiter well. The first antibiotic was loaded along the ordinate, the second antibiotic was loaded along the abscissa. The bacterial inoculum loaded was equal to 0.5 McFarland and 100 µl were distributed in the plate and were incubated for 16–24 h. The fractional inhibitory concentration indices (FICI) were calculated according to the formula: $FICI = (IC_{A+B}/IC_A) + (IC_{A+B}/IC_B)$. The interaction was considered synergistic

when FICI was ≤ 0.5 , additive when FICI was > 0.5 to ≤ 1 , antagonistic when FICI was ≥ 4 , and indifferent for intermediate value.

2.7. Time kill analysis

VIT PC 7 (CL 7) and VIT PC 9 (CL 9) were selected for time-kill analysis curves and were monitored by CLSI guidelines. Triplicates were performed for both the isolates, starting inoculum of $1-6 \times 10^6$ CFU/ml in tubes containing 20 ml of CAMH broth was used, MIC concentration of meropenem/ciprofloxacin was tested for 8th, 16th 24th hour, and then MIC was used in combination with an antibiotic, the total bactericidal log₁₀CFU/ml was determined after each hour. Bactericidal activity was defined as a ≥ 3 log₁₀ decrease in CFU/ml by 24 h, Synergy was defined as ≥ 2 log₁₀ decreases in CFU/ml at 24 h by drug

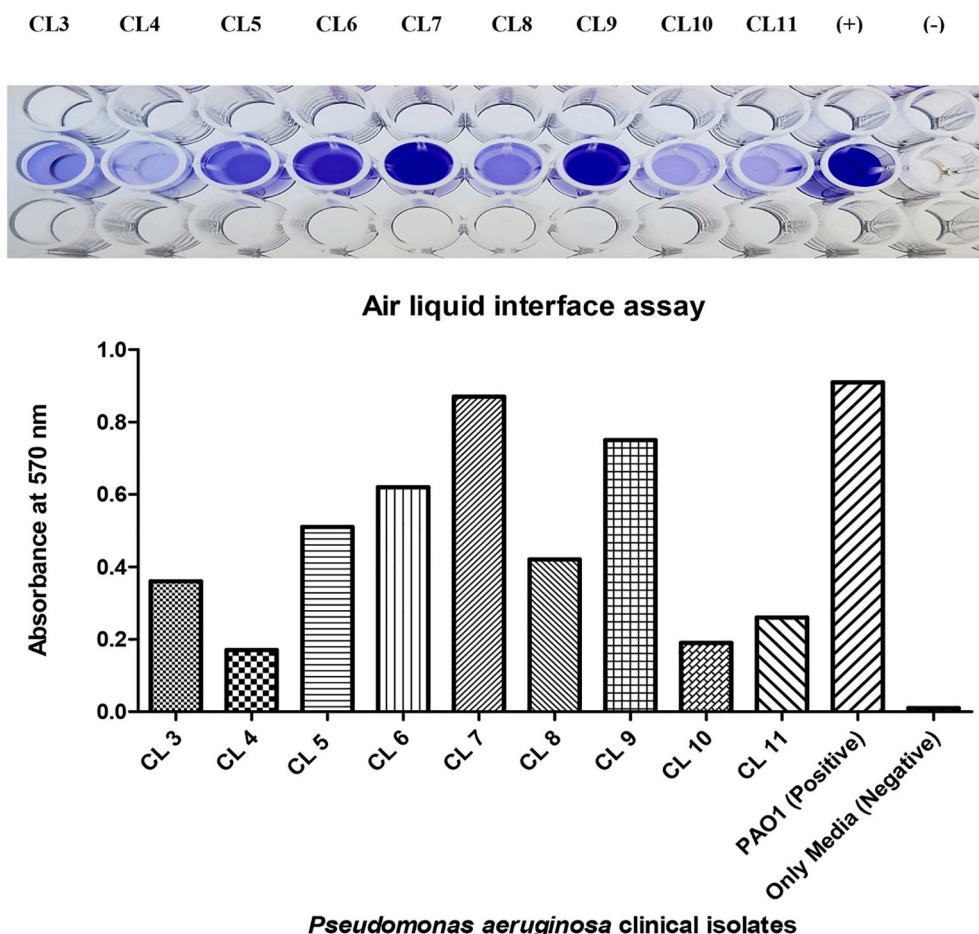


Fig. 2. Biofilm analysis using crystal violet air-liquid interface assay (Top) and at absorbance 570 nm.

Table 2

Gen bank accession number for 16sRNA sequences of *Pseudomonas* clinical isolates.

S.no	Isolate	NCBI Link
2	CL-3	https://www.ncbi.nlm.nih.gov/nucore/MK459416.1/
3	CL-4	https://www.ncbi.nlm.nih.gov/nucore/MK459417.1/
4	CL-5	https://www.ncbi.nlm.nih.gov/nucore/MK367613.1/
5	CL-6	https://www.ncbi.nlm.nih.gov/nucore/MK367614.1/
6	CL-7	https://www.ncbi.nlm.nih.gov/nucore/MK367615.1/
7	CL-8	https://www.ncbi.nlm.nih.gov/nucore/MK367618.1/
8	CL-9	https://www.ncbi.nlm.nih.gov/nucore/MK367616.1/
9	CL-10	https://www.ncbi.nlm.nih.gov/nucore/MK367617.1/
10	CL-11	https://www.ncbi.nlm.nih.gov/nucore/MK461107.1/

combination, Antagonism was defined as $\geq 2 \log_{10}$ increase in colony count at 24 h.

2.8. Combinational analysis

2.8.1. Biofilm analysis using crystal violet assay

The experiment was modified according to the analysis, VIT PC 7 and VIT PC 9 were grown overnight for 12–16 h at 37 °C. After incubation, the cultures were diluted according to 0.5 Mc Farland standard. Parallely another set of experiment was performed with the glass slides to perform SEM analysis of the test culture. For crystal violet assay, the diluted cultures were grown for 24 h in 24 well cell culture plates. After 24 h the spent media was removed and was washed twice with 1 × PBS

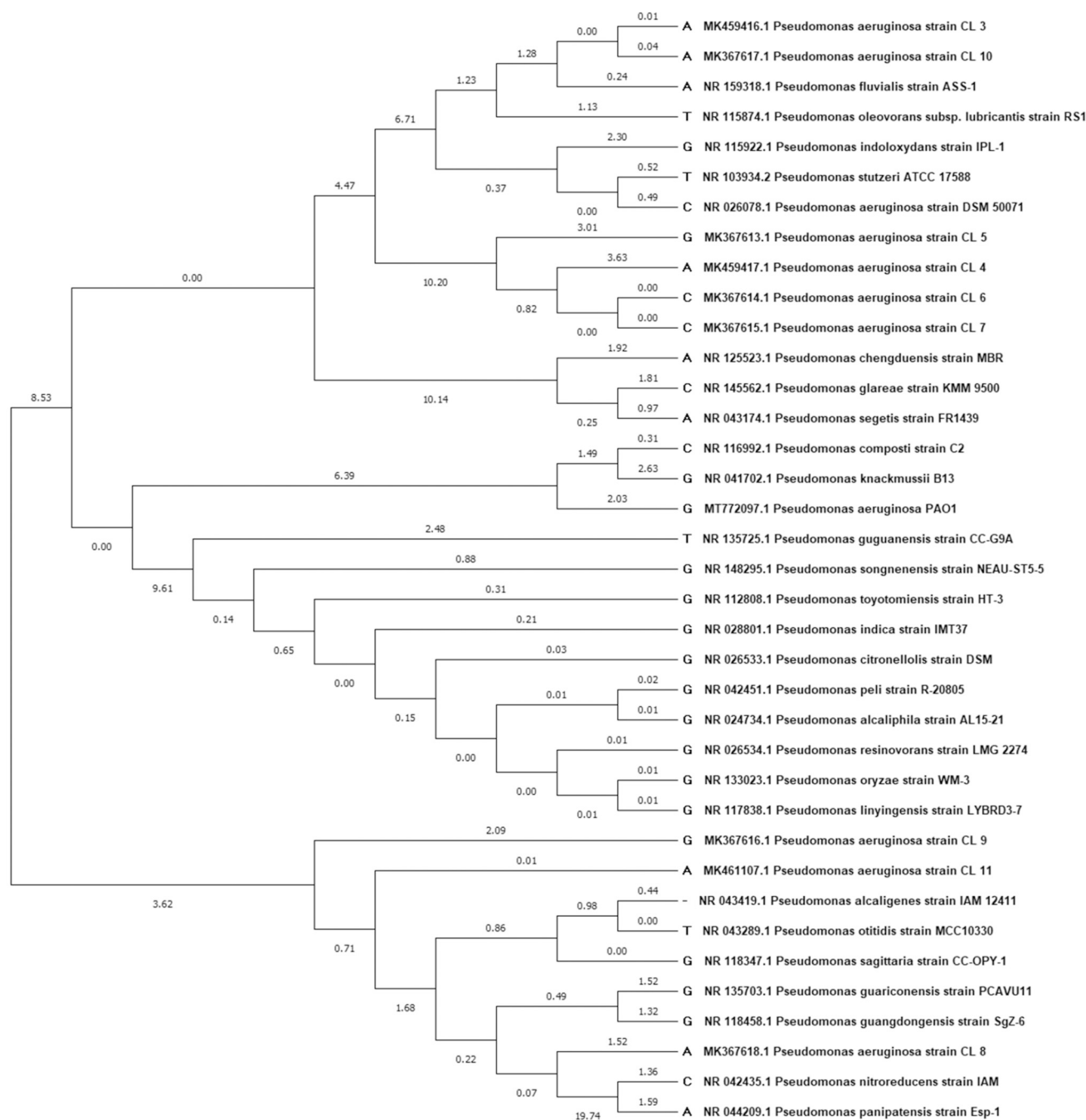


Fig. 1. Phylogenetic analysis of *Pseudomonas* clinical isolates.

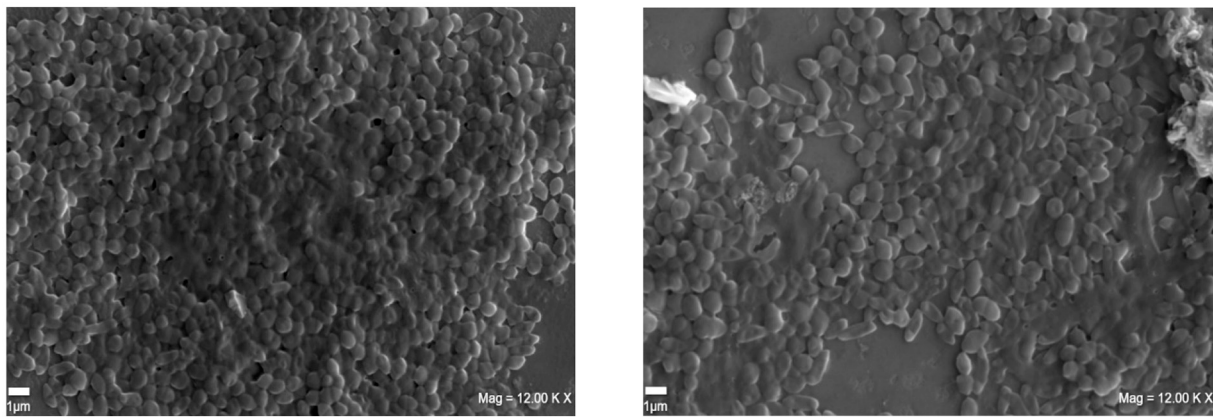


Fig. 3. SEM analysis of VIT PC 7 (left) and VIT PC 9 (right).

solution, Groups were made in the cell culture plates. Apart from control for VIT PC 7 and VIT PC 9, the test samples were as follows:

- 1) Ciprofloxacin (MIC concentration)
- 2) Meropenem (MIC concentration)
- 3) Ciprofloxacin/Meropenem (sub-MIC concentration)

The treatment was checked at the 24th hour, the wells were rinsed with $1 \times$ PBS and were air-dried, 0.1% of crystal violet solution was added at incubated for 15 mins, 30% glacial acetic acid was added to the wells and optical density was taken at O-D 570_{nm}. For SEM analysis glass pieces were added to the 24 cell culture wells (as described above) and varying treatment was performed similarly as described above, after every treatment the glass slides were fixed with 30% glutaraldehyde, dehydrated, and were sent for analysis.

2.8.2. Planktonic cells analysis

Planktonic cells were analyzed as described by et al., the 24 well plates were set as described above, after 24 h of cell growth the wells were washed with 1 XPBS and were added with the varying treatments, after every hour the spent media in the well was taken for calorimetric values, the optical density at O-D 600_{nm} and was compared with control samples.

2.9. Data analysis

Bacterial susceptibility results evaluated were classified as sensitive, intermediate, and resistant according to the CLSI testing guidelines.

3. Results

3.1. Culture characterization of *Pseudomonas* isolated from a diabetic foot ulcer

9 different *pseudomonas* isolates were collected from GVMC in the form of slants for further studies. These *Pseudomonas* isolates were further processed for confirmation of specific strain by culturing on King's B media. The colonies were identified based on fluorescent green pigmentation positive for Oxidase Test, MR-VP, and were also found to be motile (Table 1). Further, the cultures were identified based on 16sRNA sequencing, all the isolates were assigned with their accession number and were submitted to the NCBI database. 10 isolates were then subjected to multiple sequence alignment and the phylogenetic tree was deduced (Fig. 2). 16sRNA sequencing of all the 9 isolates are as follows (Table 2). (See Fig. 1.)

3.2. Biofilm detection assay (SEM analysis and crystal violet assay)

All the nine isolates were subjected for biofilm analysis and all the isolates were found to be producing biofilm, Biofilm formation was done using the crystal violet method and was subjected for 24 h incubation, Out of the 9 isolates two isolates VIT PC 7 (CL-7) and VIT PC 9 (CL-9) were found to be strong biofilm producer as compared to the rest, these 2 isolates were showing maximum production biofilm (Fig. 2.) the higher quantification value showed the potency of the XDR isolates in producing biofilm cells., SEM analysis showed the morphological characteristics showing the rod-shaped structure of isolates, microcolonies showed the surface-based growth for both the isolates (Fig. 3.) and the compact resolution of the isolates shows the potent adherence property of the cell to the substrate.

3.3. Microdilution susceptibility analysis

Both the isolates were then tested against a similar set of antibiotics to check the minimal MIC range of the antibiotics (1024-1 µg), piperacillin was found to be resistant at 1024 µg (≥ 128 µg (CLSI guidelines)) for both the isolates, showing the severity of isolate against the higher dose of antibiotic. Ciprofloxacin was found to be resistant till 64 µg of concentration for VIT PC 7 and till 256 µg for VIT PC 9 (i.e., ≥ 4 µg (CLSI guidelines)), Piperacillin/Tazobactam (Pip/Taz) was found to be resistant at 1024/4 µg (i.e., $\geq 128/4$ µg (CLSI guidelines)) for VIT PC 7 and VIT PC 9 respectively. Meropenem was found to be resistant till 128 µg and 256 µg respectively, Gentamicin was also found to be resistant at 1024 µg for both VIT PC 7 and VIT PC 9, and ceftazidime was found to be resistant against the isolate VIT PC 7 at ≥ 1024 µg concentration and VIT PC 9 respectively (Fig. 4, Table 3.).

3.4. Whole genome data analysis and resistance property

The whole-genome sequence analysis reveals yielded a draft genome for strain CL 7 with a genome size of 7,116,506 bp in 6 contigs and strain CL 9 has a genome size of 6,367,369 bp, with GC content of 65.8% and 66.4% respectively, Further, these sequences were deposited in NCBI (National Centre for Biotechnology Information) GenBank Nucleotide database as VIT PC 7 for CL-7 isolate (<https://www.ncbi.nlm.nih.gov/nucleotide/JAAGOT000000000.1>) and VIT PC 9 for CL-9 isolate (<https://www.ncbi.nlm.nih.gov/nucleotide/CP048791.1/>), comprehensive antibiotic resistance database (CARD) identified the acquired antibiotic resistance genes, and also the AMR family present in the genome of VIT PC 7 (CL 7) and VIT PC 9 (CL 9) (Tables 4a, 4b)).

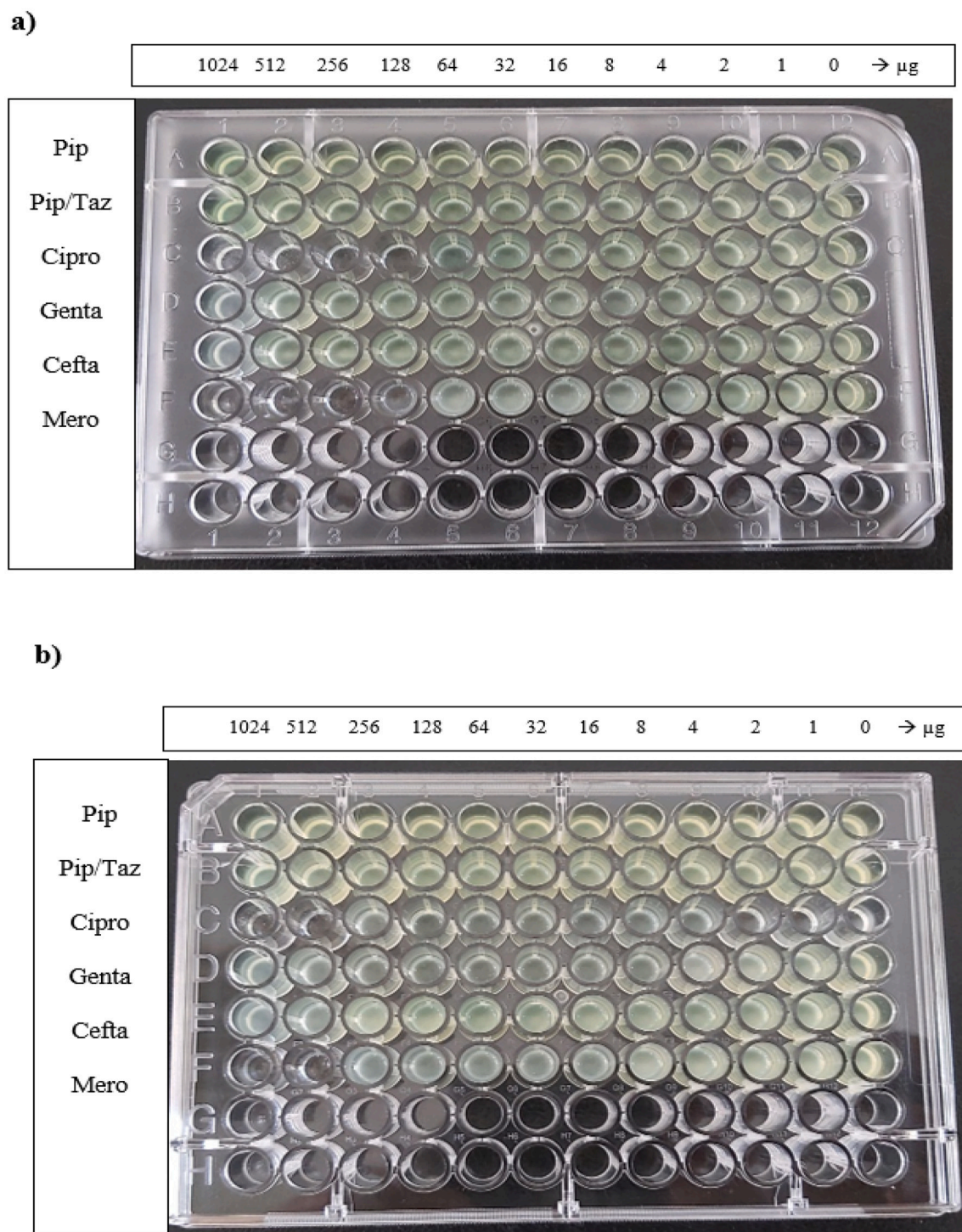


Fig. 4. Broth microdilution assay of VIT PC 7 and VIT PC 9.

Table 3

MIC value of both extensive drug resistant isolates.

S.no	Isolate	Piperacillin (MIC)	Pip/Taz (MIC)	Ciprofloxacin (MIC)	Gentamicin (MIC)	Ceftazidime (MIC)	Meropenem (MIC)
1	VIT PC 7	1024 μ g/ml	$\geq 1024/4$ μ g/ml	128 μ g/ml	≥ 1024 μ g/ml	≥ 1024 μ g/ml	256 μ g/ml
2	VIT PC 9	1024 μ g/ml	$\geq 1024/4$ μ g/ml	512 μ g/ml	≥ 1024 μ g/ml	≥ 1024 μ g/ml	512 μ g/ml

Table 4a

CARD analysis showing the presence of antibiotic resistance genes in VIT PC 7 (CL-7).

Contig	Drug Class	Resistance Mechanism	AMR Gene Family
Contig_2_759	macrolide antibiotic; fluoroquinolone antibiotic; aminoglycoside antibiotic; cephalosporin antibiotic; tetracycline antibiotic; aminocoumarin antibiotic; diaminopyrimidine antibiotic; phenicol antibiotic	antibiotic efflux	resistance- modulation -cell division (RND) antibiotic efflux pump
Contig_2_1228	aminoglycoside antibiotic	antibiotic efflux	small multidrug resistance (SMR) antibiotic efflux pump
Contig_2_1937	Cephalosporin; penam	antibiotic inactivation	OXA-beta-lactamase
Contig_2_2130	Sulphonamide antibiotic	antibiotic target replacement	sulphonamide resistant sul
Scaffold_4_33	Fluoroquinolone antibiotic	antibiotic target protection	quinolone resistance protein
Scaffold_4_69_7	bicyclomycin	antibiotic efflux	major facilitator superfamily (MFS) antibiotic efflux pump
Scaffold_4_72_4	Monobactam; carbapenem; cephalosporin	antibiotic inactivation	PDC-beta-lactamase
Scaffold_4_12_63	Macrolide antibiotic; tetracycline antibiotic; triclosan	antibiotic efflux	resistance-nodulation-cell division (RND) antibiotic efflux pump
Scaffold_4_14_10	Peptide antibiotic	antibiotic target alteration	pmrs phosphoethanolamine transferase

3.5. Combinational studies

Checkerboard analysis was done for VIT PC 7 (CL-7) and VIT PC 9 (CL-9) strain to find the additive, indifferent or synergistic effect of the antibiotic, FICI was first calculated for CL-7 for meropenem/ciprofloxacin combination as it shows prominent inhibitory effect at 256 µg and 128 µg respectively, For $\sum$ FICI, meropenem/ciprofloxacin was further checked for their least value at which there was no prominent growth, sub-MIC levels of meropenem (MIC_{1/4}) and ciprofloxacin (MIC_{1/4}) were found to be prominent in terms of synergy as it showed the $\sum$ FICI values as ≤ 0.5 (Table 5.), for VIT PC 9 meropenem showed MIC at 512 µg and ciprofloxacin showed minimum inhibition at 512 µg and in combination, it was found to be 64 µg and 128 µg respectively, which further stated the decrease in susceptibility pattern of VIT PC 9.

The result showed in Table 5, Clearly states the varying degree of fractional inhibition concentration for both extensive drug resistant isolates, For VIT PC 7 meropenem/ciprofloxacin showed some promising results when calculated at MIC_{1/4} respectively, the combinational FIC value of meropenem/ciprofloxacin showed decrease in the MIC level of the drug, But the values still comes under the range of resistance as per CLSI guidelines, The synergism activity by these two antibiotics can be further modified to some adjuvants of some combinations in order to check the efficient fractional inhibition constant for reducing the antibiotic susceptibility pattern, The extent of severity depicts the higher concentration of drug A and drug, showing the chronicity of the efflux pump associated with extensive drug resistance ((Table 4a) and (Table 4b)), The decrease in the FIC values of meropenem/ciprofloxacin

Table 4b

CARD analysis showing the presence of antibiotic resistance genes in VIT PC 9 (CL-9).

Contig	Drug Class	Resistance Mechanism	AMR Gene Family
Contig_1_281	macrolide antibiotic; fluoroquinolone antibiotic; aminoglycoside antibiotic; carbapenem; cephalosporin; cephamycin; tetracycline antibiotic; peptide antibiotic; acridine dye; aminocoumarin antibiotic; diaminopyrimidine antibiotic; sulphonamide antibiotic; phenicol antibiotic; penem	antibiotic efflux	resistance- modulation -cell division (RND) antibiotic efflux pump
Contig_1_854	Cephalosporin; penem	antibiotic inactivation	OXA-beta-lactamase
Contig_1_1543	Aminoglycoside antibiotic	Antibiotic efflux	Small multidrug resistance (SMR) antibiotic efflux pump
Contig_2_2429	phenicol antibiotic	Antibiotic inactivation	Chloramphenicol acetyltransferase (CAT)
Contig_1_2918	Fosfomycin	Antibiotic inactivation	Fosfomycin thiol transferase
Contig_1_3202	Fluoroquinolone antibiotic; aminoglycoside antibiotic; benzalkonium chloride	antibiotic efflux	Multidrug and toxic compound extrusion (MATE) transporter
Contig1_3276	Phenicol antibiotic	Antibiotic efflux	Resistance- modulation-cell division (RND) antibiotic efflux pump
Contig_1_6625	Monobactam; carbapenem; cephalosporin	Antibiotic inactivation	PDC-beta-lactamase
Contig_1_6650	Bicyclomycin	Antibiotic efflux	Major facilitator superfamily (MFS) antibiotic efflux pump
Contig_1_6738	Fluoroquinolone antibiotic; tetracycline antibiotic; acridine dye	Antibiotic efflux	Resistance- modulation-cell division (RND) antibiotic efflux pump

depicts that as required range for the isolate to be sensitive to meropenem/ciprofloxacin is [≤ 2 µg (susceptible), 4 µg (intermediate), ≥ 8 µg (resistant)], for ciprofloxacin (≤ 1 µg (susceptible), 2 µg (intermediate), ≥ 4 µg (resistant))] but the decrease in concentration from single MIC value paves way to monitor the use of varying concentration of antibiotics in combinations. These reductions in MIC values may have a greater impact when dealing with the chronic condition of extensively drug-resistant *Pseudomonas sp* as these values shows the act of synergism of antibiotics when used in different concentration, The combination of meropenem/ciprofloxacin at their MIC value and their sub-MIC (synergistic value) was evaluated for Time kill analysis, The VIT PC 7 and VIT PC 9 showed the significant decrease in the number of viable cells at 24th hours when checked at their synergistic MIC concentration, No antagonistic effect was seen showing the potency of the drug at their combined effect at MIC value Fig. 5a.

Similarly, VIT PC 9 isolate also showed the synergistic effect at MIC showing the drastic reduction in the number of CFU count, (Fig. 5b).

Table 5

Fractional inhibitory concentrations of VIT PC 7 and VIT PC 9 (Light grey shows additive effect 0.5–1, medium grey shows synergistic effect ≤ 0.5 and dark grey shows indifferent <1).

Isolate	Antibiotic (A) in $\mu\text{g/ml}$	Antibiotic (B) $\mu\text{g/ml}$	MIC of drug (A)	MIC of drug (B)	FIC of drug (A) in combination	FIC of drug (B) in combination	$\sum\text{FIC}$ $\sum\text{FICI} = \text{FIC of drug A and FIC of drug B}$
VIT PC 7	Meropenem	Ciprofloxacin	256 μg	128 μg	64 μg	32 μg	Synergistic
VIT PC 9	Meropenem	Ciprofloxacin	512 μg	512 μg	128 μg	64 μg	Synergistic

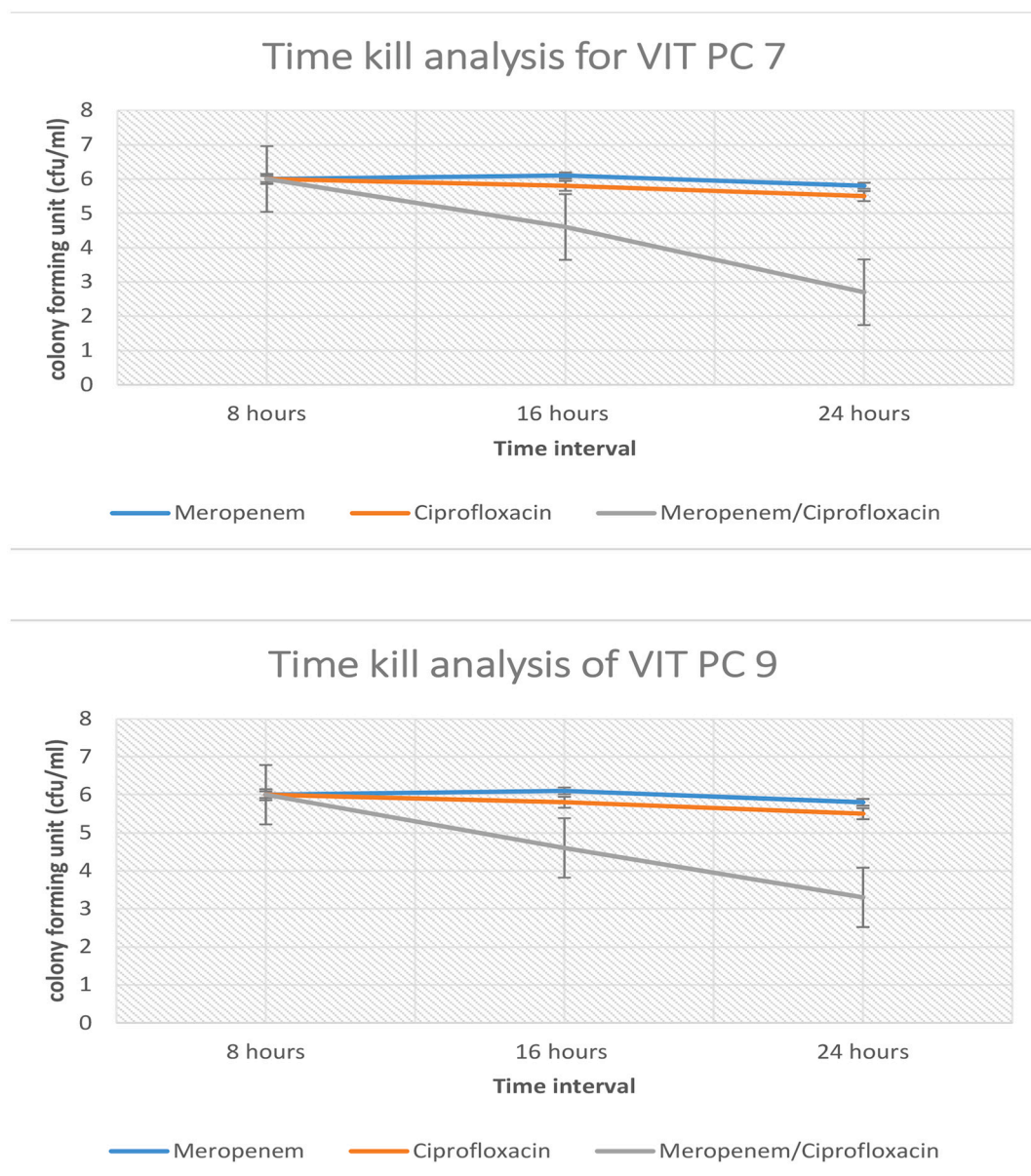


Fig. 5. a. Time kill analysis of VIT PC 7 showing synergy at sub-MIC of Mer/Cipro combination. b. Time kill analysis of VIT PC 9 showing synergy at sub-MIC of Mer/Cipro combination.

3.6. Activity of synergistic combination against preformed biofilm of *Pseudomonas* isolates

sub-MIC level of ciprofloxacin/Meropenem was used to check the effect of combination against isolates, crystal violet assay showed maximum reduction in absorbance as compared to the individual MIC

assay of antibiotics, the optical density of the isolates shows the inhibitory activity against preformed biofilm, 24 h treated sample was then analyzed for planktonic cell assay, absorbance at 600 nm showed the reduction in even planktonic cells which further infer the inhibitory activity of the synergistic dose of the drug against the free-flowing cells (Fig. 6a, b).

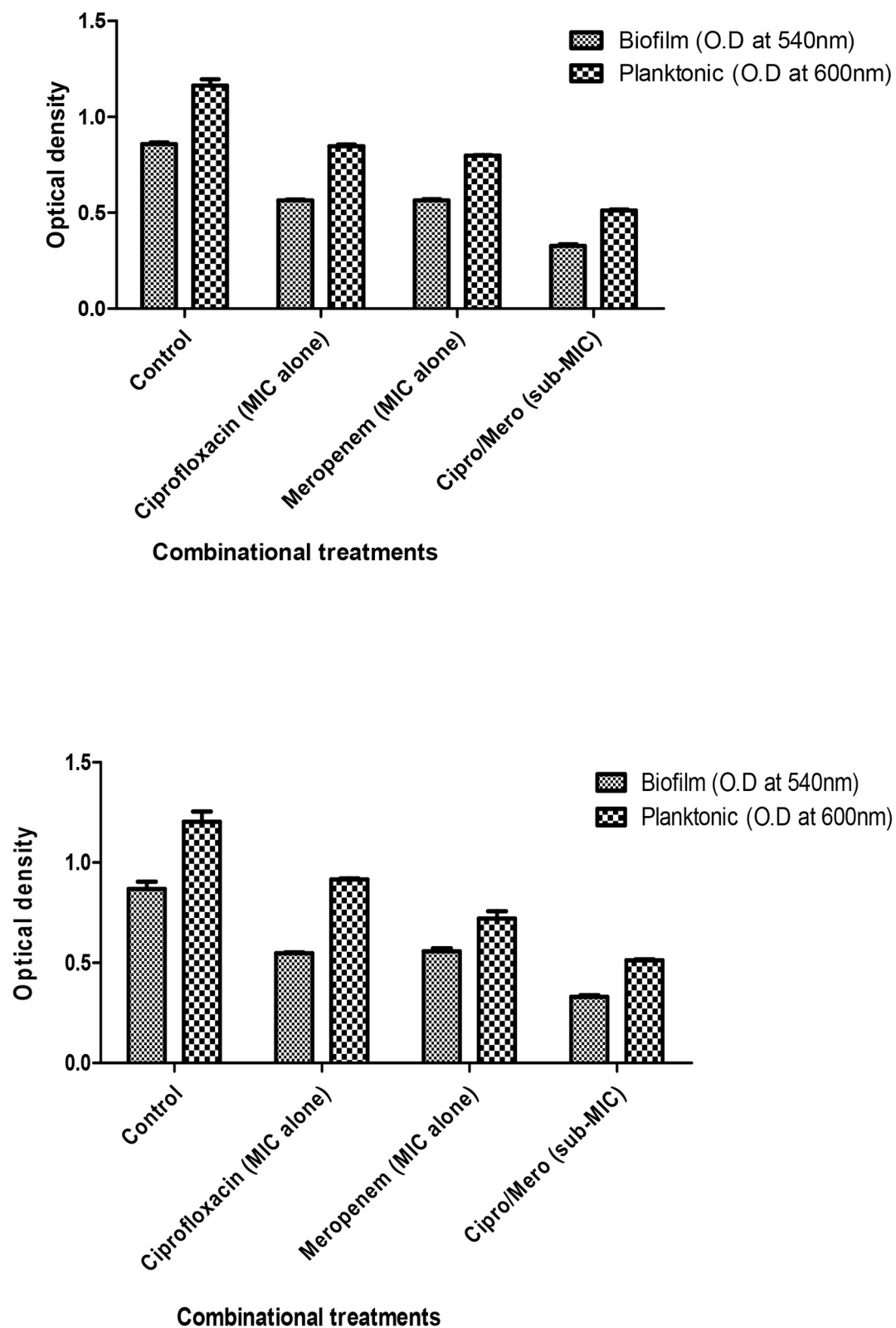


Fig. 6. a. Combinational treatment (biofilm/planktonic cells) on VIT PC 7. b. Combinational treatment (biofilm/planktonic cells) on VIT PC 9.

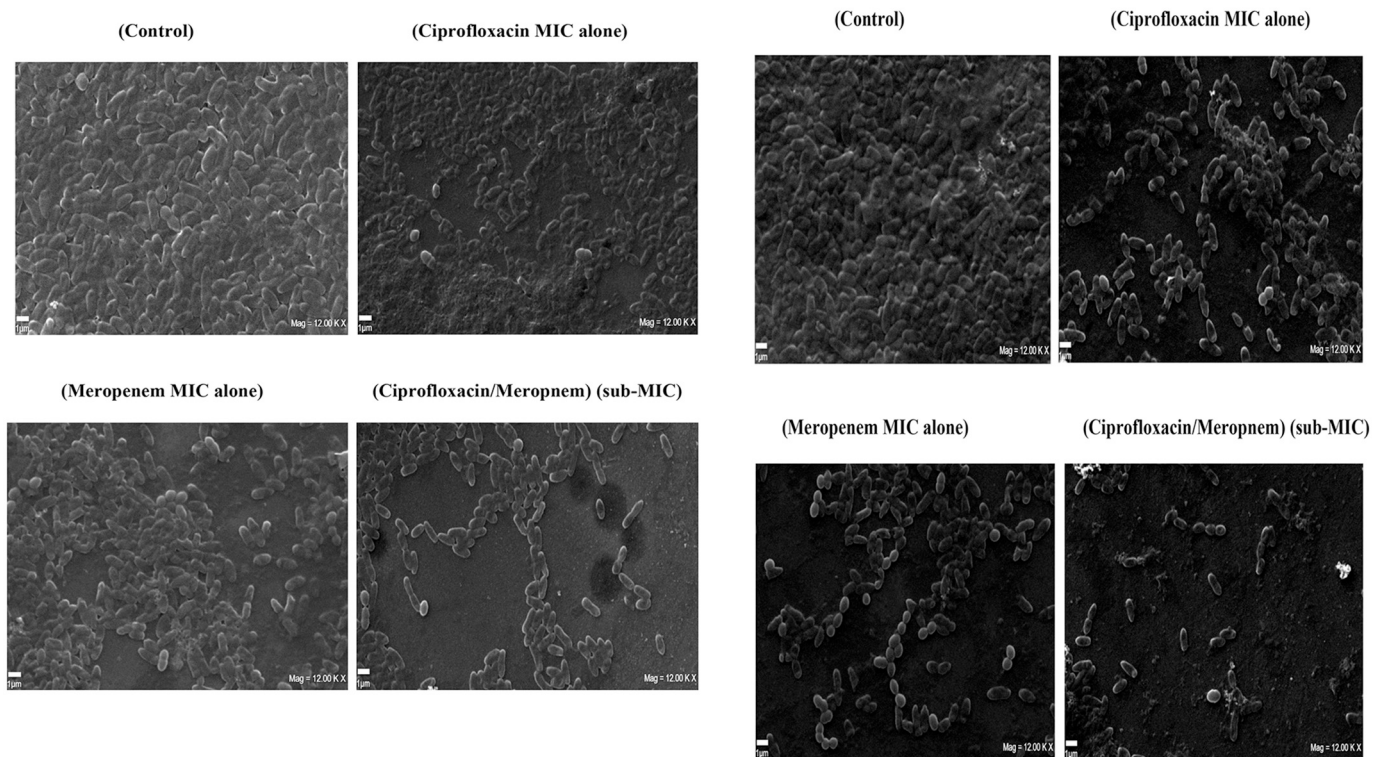


Fig. 7. a. SEM analysis of combinational treatment against VIT PC 7. b. SEM analysis of combinational treatment against VIT PC 9.

SEM analysis also showed the reduction in the number of biofilm-forming cells portraying the antibiofilm activity of the combinational dose of the drug, these results further infer the antibacterial and antibiofilm activity of the sub-MIC level of ciprofloxacin/meropenem and can be further analyzed at the transcriptomic level to check the interaction against various drug-resistant pumps related to ciprofloxacin/meropenem (Fig. 7a, b).

4. Discussion

90% of individuals with diabetic foot issues and lower extremity amputation have stories of foot ulcers, which is the leading cause for morbidity or mortality upon long time hospitalization (Rees et al., 2018). Exposure of these wounds to microbiota results in the habitation of pathogenic and non-pathogenic microbes for days to weeks. Predominantly *Staphylococcus aureus* >30%, *Pseudomonas aeruginosa* >15%, *Proteus* >19%, *E. coli* and *Corynebacterium* >5% are reported to be the potent pathogens, infecting these wounds. These aerobic and anaerobic microbes due to their synergistic infections such as gangrene, necrosis makes diabetic foot ulcer patients' life much more complicated.

Monotherapy against XDR *Pseudomonas* doesn't give appropriate results as they have lost the potency against serious *Pseudomonas* infections, the use of double carbapenem have also been proven to be ineffective against monotherapy of carbapenem (American and Hoşgör, 2001). WHO itself has put *Pseudomonas* infection in the most critical infection along with *Acinetobacter* and *Enterobacteriaceae* (Isenberg et al., 1999). Due to extensive use of ciprofloxacin antibiotic it has been seen that antibiotic has been resistant to *Pseudomonas* infection (Hidayat et al., 2006). The increased rate of mutation in pathogen makes existing antibiotic-resistant (Idowu et al., 2019). So in order to increase the potency of antibiotic, Fluoroquinolones can be used in combination of other anti-pseudo antibiotics, Ciprofloxacin studies have been done previously in combination of β -Lactam antibiotics, and it has shown some significant reduction in the MIC count of infection, Study conducted by (Nichols et al., 2016) showed the increased efficiency of

ciprofloxacin/imipenem against ciprofloxacin/amikacin combination showing the effective synergism against *Pseudomonas* infection, Liposomal mediated core delivery of colistin and ciprofloxacin showed synergistic effect against XDR *Pseudomonas aeruginosa* (Torres et al., 2012).

Multiple drug resistance (MDR) has emerged as a major threat to hospitalized and hospital associated infections, leading to mortality ranging from 30 to 70%. Mostly it is observed that this is due to inappropriate usage of broad-spectrum antibiotics which leads to multiple drug resistance of *Pseudomonas*. To overcome this MDR property and to reduce the impact of diseases caused by MDR strains, combinational therapy is used as an alternative and effective strategy to fight the resistance (Srivastava and Sivashanmugam, 2020). To explore combinational therapy and its synergic activity Σ FICI (fractional inhibitory concentration) using checkerboard method and time kill analysis of ciprofloxacin/meropenem were carried out in our experiment, this combination of two different antibiotic therapy consisting of β -Lactam and fluoroquinolone compounds inhibiting *Pseudomonas* strains VIT PC 7 and VIT PC 9 has given prominent fold reduction. It was observed that 4-fold and 8-fold increase in MIC (minimum inhibitory concentration) was observed for the strains VIT PC 7 and VIT PC 9 respectively, showing maximum synergistic effect of the two antibiotics. Overall, both the combinational treatment has found to suppress biofilm formation of VIT PC 7 and VIT PC 9 which were proven using SEM (scanning electron microscope) and air liquid interface assay (crystal violet assay). Since we have found the presence of multiple drug resistance to various antibiotics for both VIT PC 7 and VIT PC 9 (Srivastava et al., 2020), further studies on gene expression and transcriptome will help us to unravel further impact of synergistic and combinatorial antibiotic therapy.

5. Conclusion

In our study, we have isolated 2 most potent extensive drug-resistant strains of *Pseudomonas aeruginosa*, VIT PC 7 and VIT PC 9 from diabetic foot ulcer patients of GVMCH, Vellore and have studied their various

resistance pattern using whole-genome sequencing [44], susceptibility studies were performed using broth microdilution assay showing the effect of meropenem/ciprofloxacin susceptibility at higher concentration, paving the way to design combinational drug analysis against these extensive drug resistance strains. To reduce the drug resistance pattern of these two isolates VIT PC 7 and VIT PC 9, we have carried out synergistic approach and checkerboard analysis which would enhance the chance of lowering the MIC value of these drugs against these two isolates. The drug penetration was observed to be significantly better when meropenem was used in combination with ciprofloxacin against VIT PC 7 and VIT PC 9 showing the enhanced drug susceptibility by 4-fold and 8-fold but still, the range of the antibiotic showed the resistance mechanism showing the potency of the antibiotic resistance bacteria, further we are planning to explore the upregulation and downregulation of the biofilm-forming genes and multiple drug-resistant genes in using transcriptome analysis.

Ethics statement

The Institutional Ethical Committee approval certificate was obtained from Government Vellore Medical Hospital (GVMC), Vellore, Tamil Nadu for the collection of pus samples from diabetic foot ulcer patients. The Institutional Ethical Committee (IEC) meeting was held on 06-10-2016 at conference hall, (GVMC), Vellore. The members of the committee, secretary, convenor, and the president approved the proposed work entitled "In vitro and In vivo studies of the inhibitory effect of *Bacillus* derived N-acyl Homoserine Lactonase degrading enzyme (Aiia) on *Pseudomonas aeruginosa* in Diabetic Foot Ulcer".

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships which have, or could be perceived to have, influenced the work reported in this article.

I confirm that this work is original and has not been published elsewhere nor is it currently under consideration for publication elsewhere and we have no conflicts of interest to disclose.

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